Faculty Initial: GM

Dept. of CSE

School of Science and Engineering

Date: Dec 27, 2024

Examination	: Mid Examination	**Semester**	: Fall 2024
Course Code	: MAT141	**Marks**	: 20
Course Title	: Differential & Integral Calculus	**Section**	: 8, 11, 12
Slot	: Slot - 22	**Duration**	: 1 hour 30 minutes

General Instructions:

- All questions must be answered.
- All the devices (mobile, smart watch, etc.) should be turned off and kept in the student's bag during the exam period.
- The question paper should be submitted during the answer script submission.
- In questions, $a, b, c, d, e, m, g,$ and h are arbitrary constants related to the student ID. For example, for the ID-2023200000790, use the format $202ab00000cde$. Thus $a = 3, b = 2, c = 7, d = 9, e = 0, m = c + d + e = 7 + 9 + 0 = 16$, $g = a * e + 1 = 3 * 0 + 1 = 1$ and $h = b + c + 2 = 2 + 7 + 2 = 11$.

Sl. No.	Tasks	COs	BCL*	Marks
1.	Compute the limit of $f(x) = \frac{\lvert x-m \rvert}{x-m}$ at $x = m$ (if exist).	CO1	L-2 [Understand]	2
2.	If $f(x) = \begin{cases} x^2+1, & x < h \\ x, & h \le x \le h+1 \\ \frac{1}{x}, & x > h+1 \end{cases}$, then discuss the continuity and differentiability of $f(x)$ at $x = h$.	CO1	L-2 [Understand]	3 + 2
3.	Calculate the intervals in which the function $f(x) = 8x^3 - 60x^2 + 144x + 15$ is increasing or decreasing.	CO1	L-3 [Apply]	2
4.	Construct the differential co-efficient of the function $y = sinax + e^{bx} + 5x^g + cx^3y^2 + msin^{-1}x$ with respect to x.	CO2	L-3 [Apply]	3
5.	Differentiate the following functions: (i) $mx^2 + g$ by using the first principle. (2 marks) (ii) x^{sinhx} with respect to $(cosax)^{cx}$. (3 marks)	CO2	L-4 [Analyze]	5
6.	A manufacturer finds that the marginal cost (in dollars) associated with the production of x units of a photocopier component is given by $30 - 0.02x$. If the cost of producing one unit is \$35. Compute the cost function and the cost of producing 100 units.	CO4	L-3 [Apply]	3

***Bloom's Cognitive Level**